Module – 3 & 4

Assignment Submissions – embedUR Networking Course

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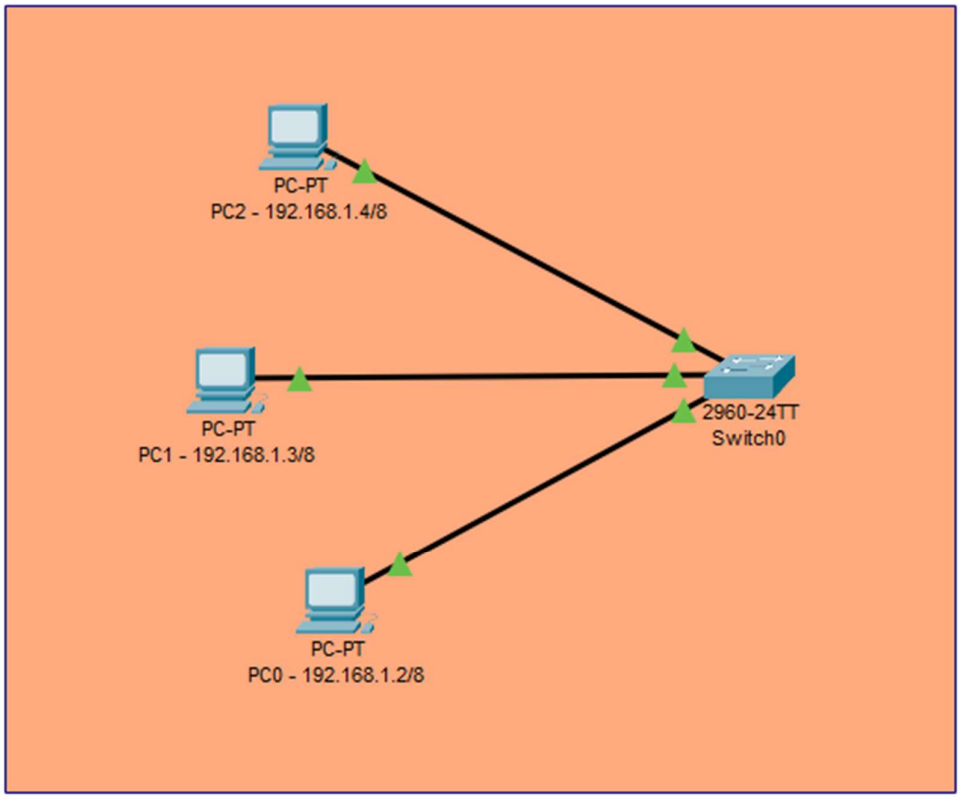
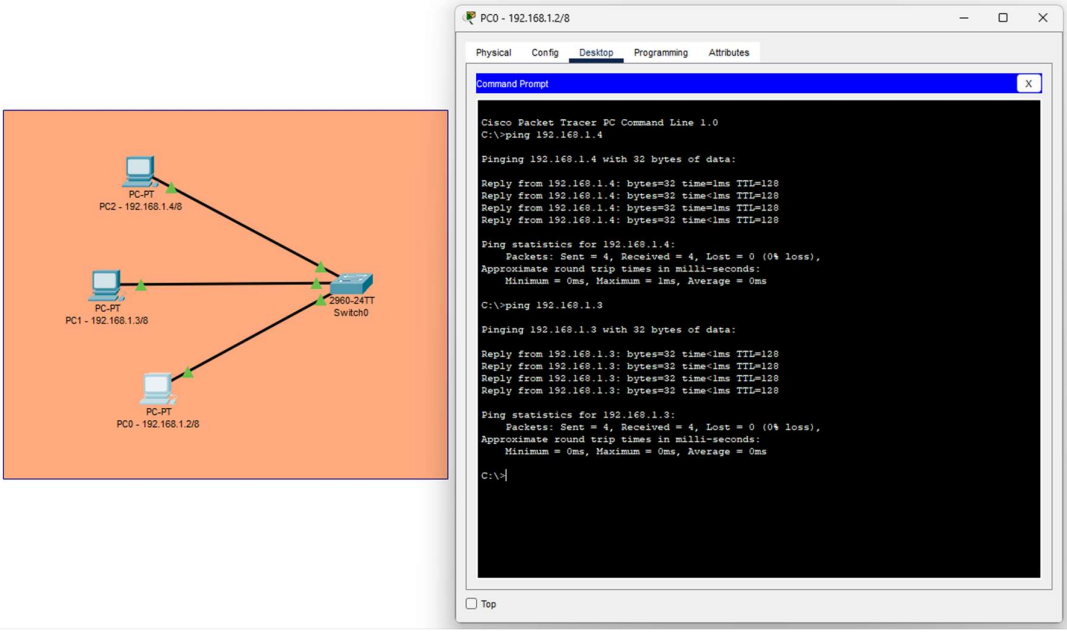
Simulate a small network with switches and multiple devices. Use ping to generate traffic and observe the MAC address table of the switch. Capture packets using Wireshark to analyze Ethernet frames and MAC addressing.

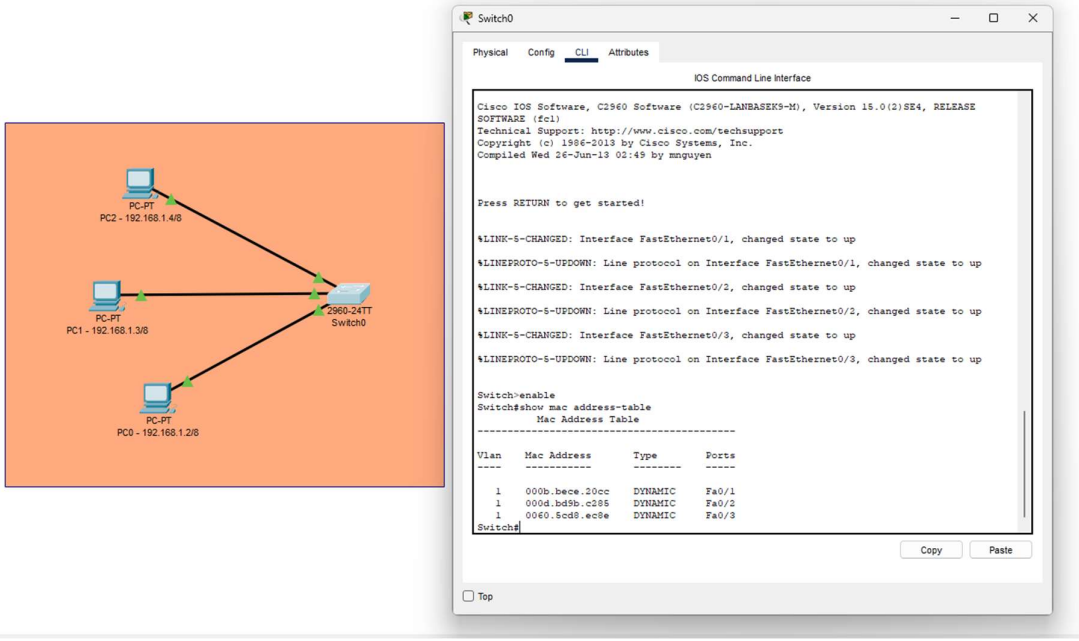
Capture and analyze Ethernet frames using Wireshark. Inspect the structure of the frame, including destination and source MAC addresses, Ethertype, payload, and FCS. Use GNS3 or Packet Tracer to simulate network traffic.



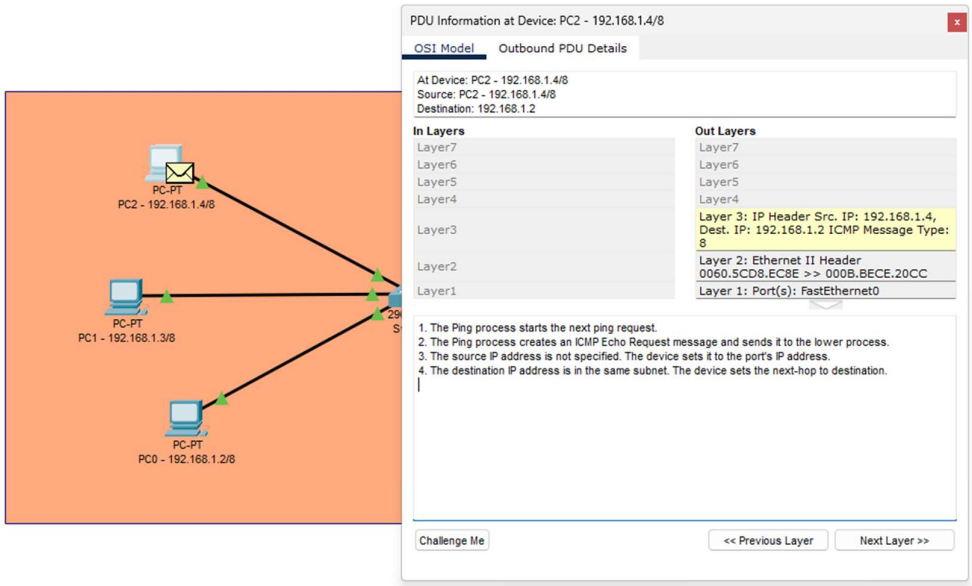
Sample Network to be used for assignment



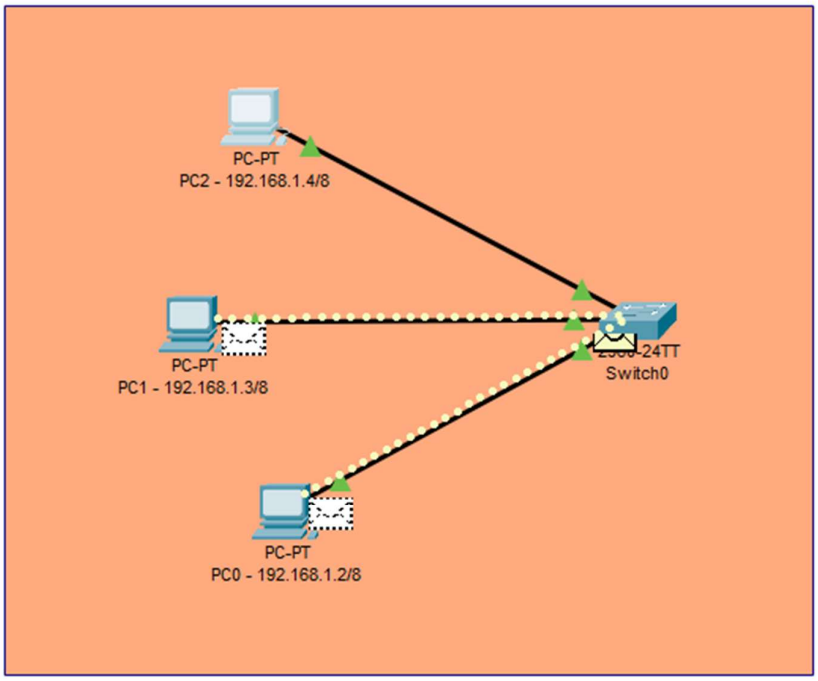
Pinging PC2 from PC0

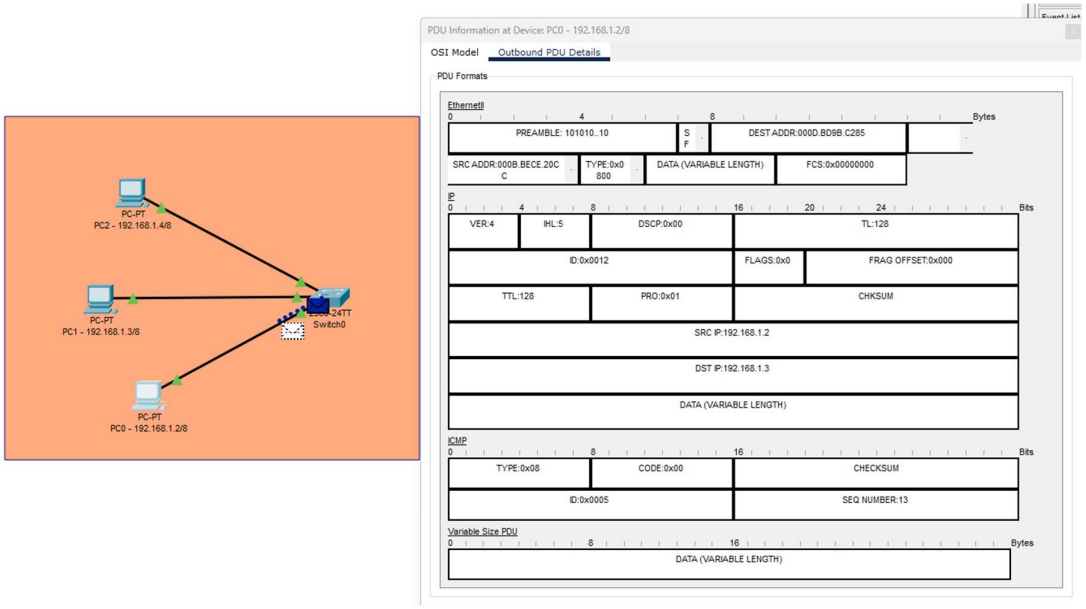


Mac Address table from Switch



PDU details on ping from PC2 to PC0

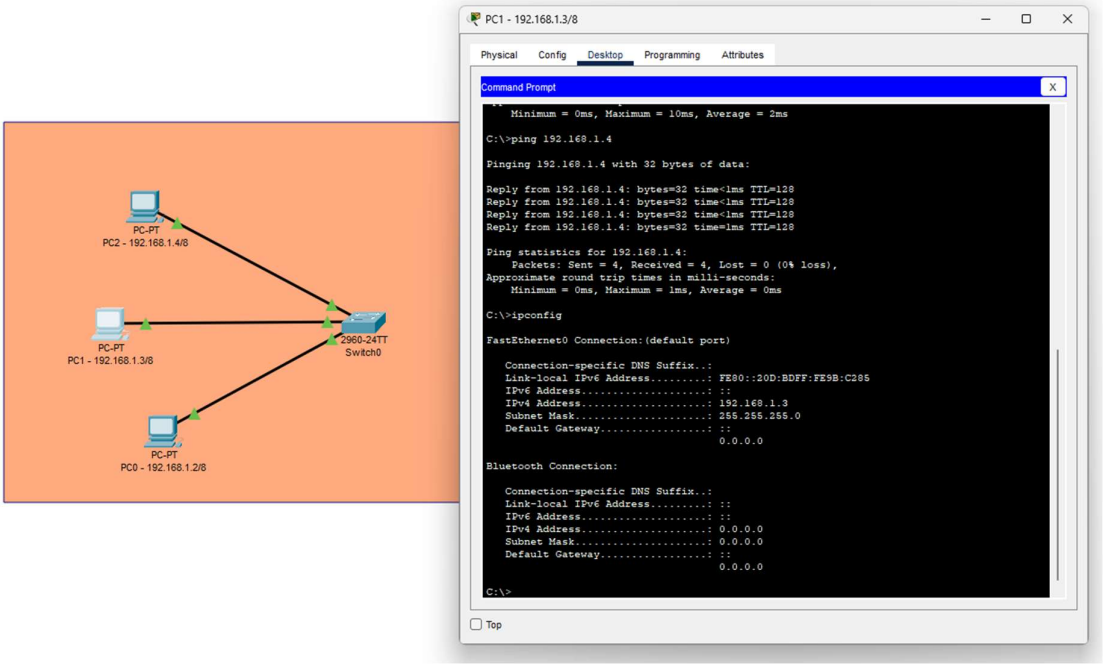




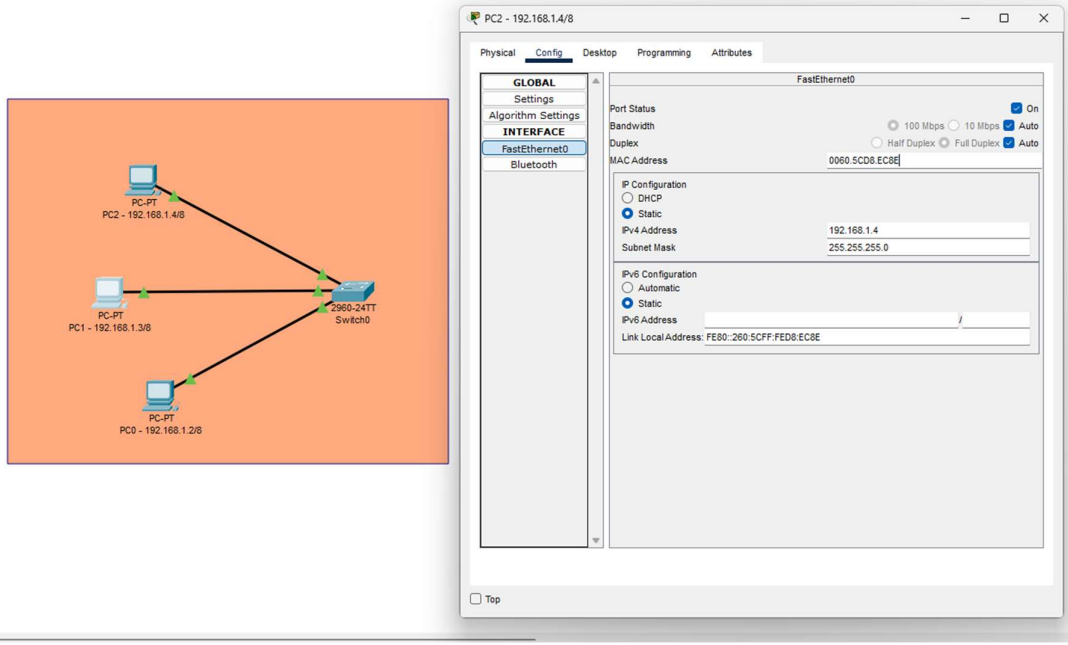
Ethernet Frame results

Configure static IP addresses, modify MAC addresses, and verify network connectivity using ping and ifconfig commands.

Testing the network config – using the ipconfig and ping commands

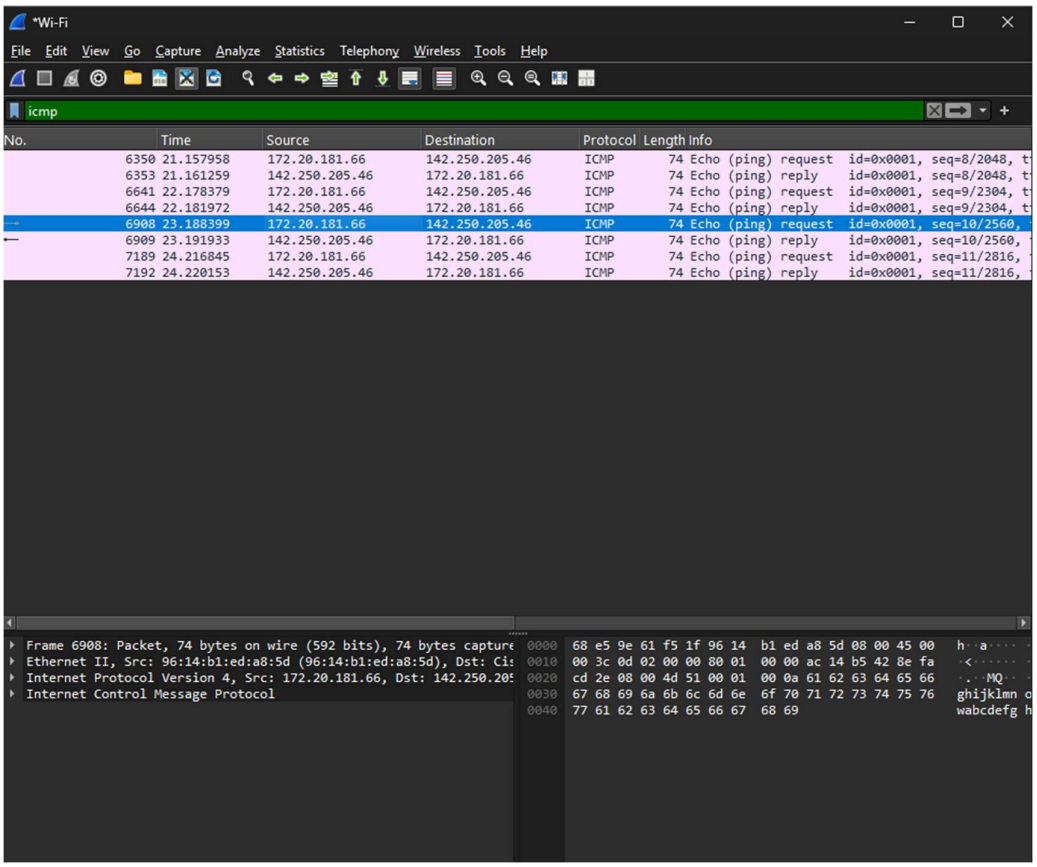


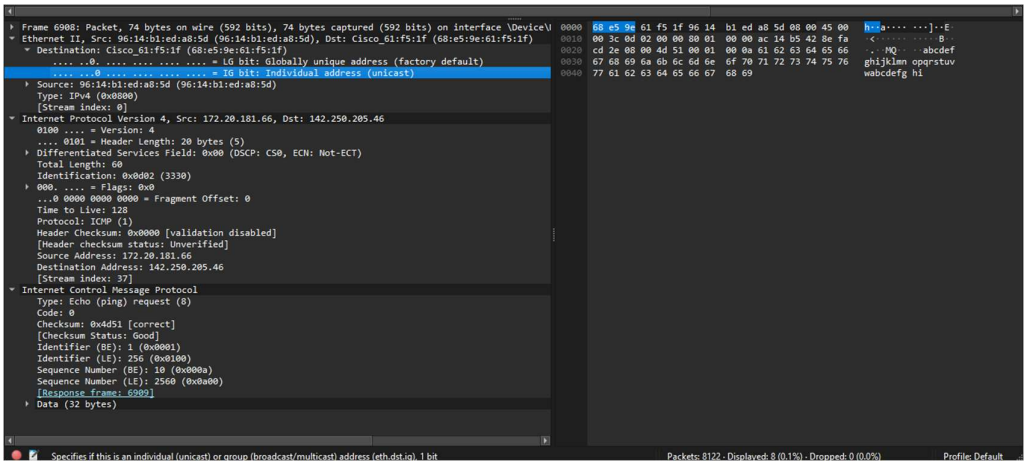
And changed MAC address of PC1



Capture and analyze Ethernet frames using Wireshark. Inspect the structure of the frame, including destination and source MAC addresses, Ethertype, payload, and FCS. Use GNS3 or Packet Tracer to simulate network traffic.

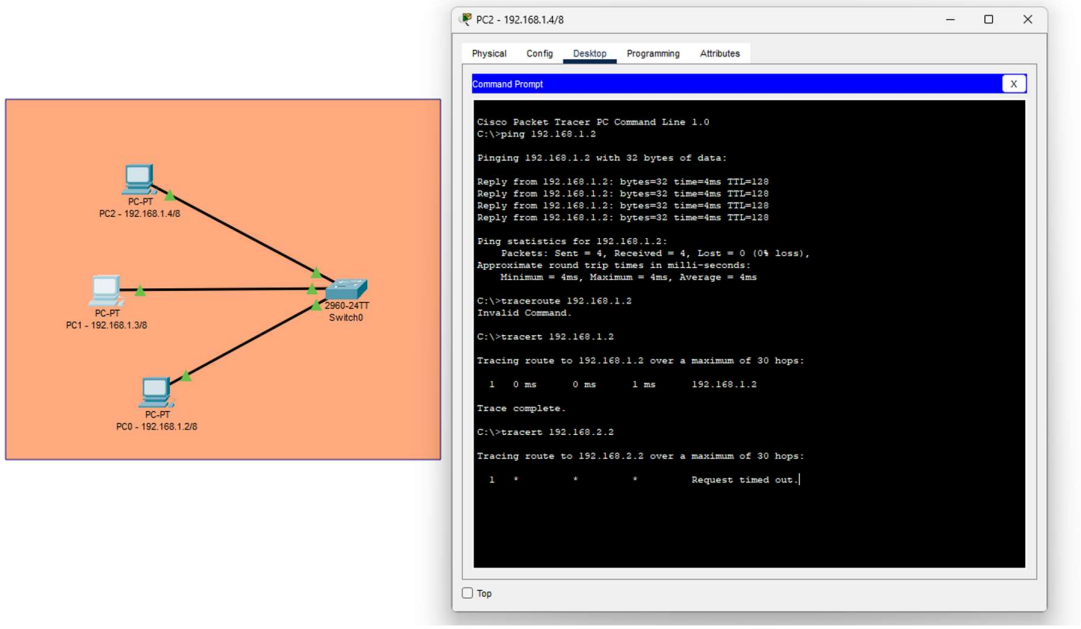
Filtered ICMP packets (ping google.com command)





Troubleshoot Ethernet Communication with ping and traceroute -> Using cisco packet tracer:

- **Create a simple LAN setup with two Linux machines connected via a switch.**
- **Ping from one machine to the other. If it fails, use ifconfig to ensure the IP addresses are configured correctly.**
- **Use traceroute to identify where the packets are being dropped if the ping fails.**



Research the Linux kernel's handling of Ethernet devices and network interfaces. Write a short report on how the Linux kernel supports Ethernet communication (referencing kernel.org documentation).

The Linux kernel manages Ethernet communication through network device drivers and the network interface layer.

Key components:

- **net_device structure** : Core data structure in the kernel that represents every network interface (like eth0). It holds MAC address, MTU, and interface state.
- **NAPI (New API)** : Interrupt-driven packet processing mechanism used by the kernel to handle high-speed Ethernet traffic efficiently.
- **Socket Buffer (sk_buff)** : Data structure used to hold packet data as it moves through the kernel's network stack.
- **Ethtool** : Kernel interface that allows querying and configuring Ethernet device parameters.

The kernel handles Ethernet frames by passing them from the NIC driver → network stack → socket layer → user application.

Referred from : https://www.kernel.org/doc/html/latest/networking/netdevices.html

Configuring a Basic LAN Interface using the ip command

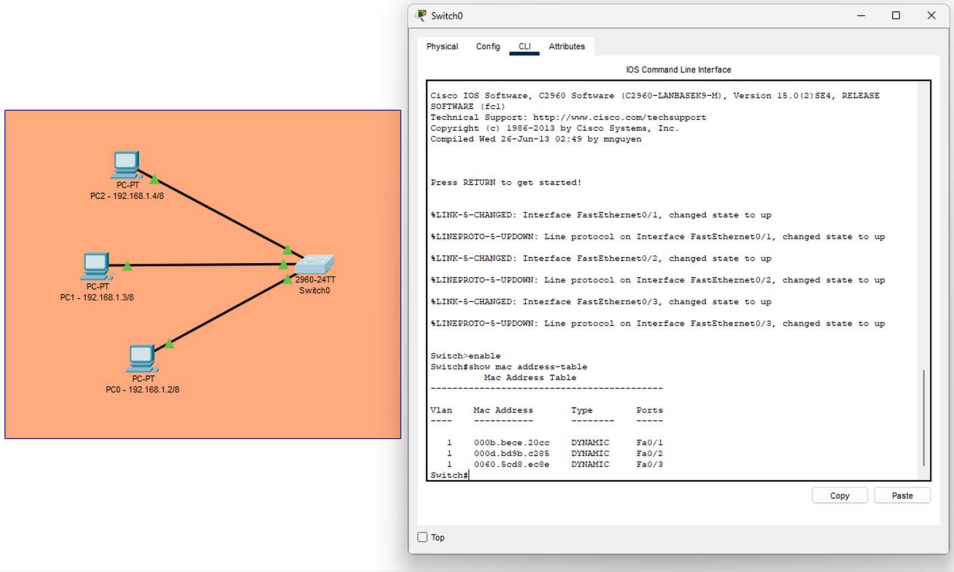
Commands used:

ip link set eth0 up

ip addr add 192.168.1.10/24 dev eth0

ip route add default via 192.168.1.1

ip addr show eth0

- ip link set eth0 up : Brings the network interface up
- ip addr add : Assigns a static IP address to the interface
- ip route add default via : Sets the default gateway
- ip addr show : Verifies the configuration applied correctly

Use Linux to view the MAC address table of a switch (if using a Linux-based network switch). Use the bridge or ip link commands to inspect the MAC table and demonstrate a basic switch's operation.